

Exploring the Synergy of Combining eXtended Reality and a Robotic Arm for Supporting Scenarios of Remote Collaboration

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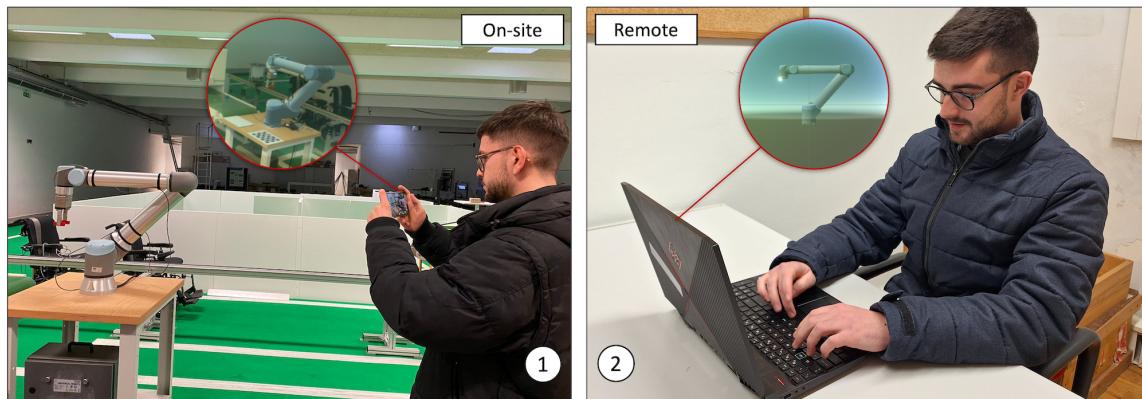


Figure 1: Scenario of remote collaboration supported by XR, involving (1) a robotic arm, an on-site member supported by a mobile device and (2) a remote member using a virtual replica in a computer to provide assistance.

ABSTRACT

The growing interest in remote collaboration arises from different factors, most notably the rapid advancements in digital technologies, making it increasingly feasible to work from any location. These innovations open new possibilities for global cooperation and productivity. However, several challenges hinder seamless collaboration. Among these are the inherent limitations faced by remote participants in interacting directly with on-site physical objects. Additionally, their perspective remains significantly constrained, often limited to the viewpoints and visual inputs shared by on-site members. Addressing these is essential to unlock the full potential of remote collaboration. This paper explores a robotic arm within an eXtended Reality (XR) environment to transform collaboration practices. The architecture and main features are described, including streaming the task context, on-site and remote manipulation of the robotic arm, and the capacity to display safety zones, among others. Finally, a reflection on applicability is discussed and future directions are drawn.

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1 INTRODUCTION

Remote collaboration using eXtended Reality (XR) has gained significant prominence across a wide range of domains in recent years [5, 15, 19, 23]. In industries, it facilitates efficient project management and innovation by enabling experts from different locations to pool their knowledge and skills. In healthcare, remote collaboration has transformed patient care through telemedicine, allowing specialists to consult and perform diagnostics across distances. Similarly, in education and training, it has created new opportunities for inclusive and flexible learning experiences, connecting students and professionals with resources and expertise worldwide [9, 20, 31, 32]. As technology continues to advance, remote collaboration undergoes constant evolution, paving the way for increasingly seamless, effective, and inclusive ways of working together. Innovations in digital tools, communication platforms, and interactive technologies enable individuals and teams to connect and collaborate in real-time, overcoming barriers of distance and time zones [9, 28, 36, 37]. Using XR also create opportunities for more equitable participation, ensuring that diverse voices and expertise can contribute to shared goals. The ongoing transformation of remote collaboration is reshaping how people interact, innovate, and achieve success in a connected world. By utilizing immersive technologies to transcend geographical barriers, digital technologies like eXtended Reality (XR) can create interactive and immersive environments that redefine collaboration possibilities in ways that were once constrained by geographical boundaries [6, 21, 26, 29]. These solutions can establish a common ground for situation understanding, idea discussion, and instruction generation, creating symbioses between the real and virtual worlds [4, 12, 18, 24, 25].

Yet, XR still encounters limitations when facing remote scenarios. Regarding **limited perspective and context capture**, the remote

member's view is confined to what on-site members can capture, potentially hindering comprehensive environmental understanding and impeding effective remote guidance and decision-making. Likewise, XR-based solutions mainly focus on image/video sharing, suggesting a **lack of multi-sensory data collection** that inhibits remote members from gaining a nuanced and thorough understanding of the physical surroundings. This deprives remote collaborators of other essential/contextual cues, hindering their ability to better assess the intricacies of the on-site environment. Additionally, concerning **inadequate physical interaction**, XR's interaction with on-site physical objects may lack the depth required for detailed explanations or nuanced gestures in complex or dynamic scenarios. Certain collaborative tasks necessitate the precision and dexterity that using XR-based solutions alone may struggle to provide. As for **navigational challenges and safety**, on-site members using XR may find it difficult to maneuver through complex or hazardous environments. This difficulty in navigation can impede context sharing, as well as the efficiency and effectiveness of their tasks [7, 13, 33, 34, 38].

This work tries to address some of these challenges by integrating a static robotic arm into an XR environment, aiming to redefine how such scenarios occur. Robots can be dynamic agents, acting as the eyes and ears of remote collaborators and extending beyond fixed perspectives. Robots may offer enhanced mobility for comprehensive environmental exploration, providing additional angles and viewpoints beyond the on-site member's perspective. Hence, providing on-site members the freedom to engage in parallel tasks, unburdened by the need to concentrate on capturing the context of the ongoing task. [2, 7, 8, 22]. Moreover, these robots can navigate challenging terrains or hazardous environments where human access is restricted, while also facilitating interaction through advanced sensors for manipulating physical objects at a distance. This interactivity effectively bridges the gap between physical and virtual realms, empowering team members with enhanced capabilities and fostering more effective communication [3, 11, 27, 30].

The proposed prototype facilitates collaboration by enabling an on-site member, equipped with a handheld device, to capture the task context and share it with a remote counterpart through a live stream. This can also be conducted through a dedicated camera attached to the robotic arm. The remote member, using a computer, can analyze the situation and provide real-time assistance by controlling the robotic arm through a specialized interface. Additionally, the system allows for the definition of safety zones, ensuring that on-site members can operate safely within the environment. These features, alongside others, are described, showcasing its versatility and potential to assist with remote practices.

2 XR AND A ROBOTIC ARM FOR REMOTE SCENARIOS

To start addressing some of the aforementioned challenges, a first effort has been made, which will be presented in the form of a platform using an XR environment and a static robotic arm. The following scenario was considered (see Figure 1): an assembly task performed collaboratively by an on-site member and a robotic arm, with support from a remote member located in a distinct setting. Such tasks are commonly used in collaborative scenarios as an initial use case to identify key requirements and guide the design and development of effective collaborative tools. By simulating real-world interactions and challenges, this scenario serves as a foundational framework to refine and validate the system's capabilities in addressing the complexities of remote collaboration.

As for deployment, it begins with the placement of a pre-defined marker in close proximity to the robotic arm. This step is essential to ensure that the augmented content, including visual overlays and interactive elements, is accurately aligned with the physical robotic arm. Proper alignment facilitates seamless integration between the virtual and physical components, enabling precise interactions and enhancing the overall effectiveness of the system during collabo-

orative tasks. Afterward, when the on-site member mobile device's camera detects it, an array of XR interfaces is triggered, providing real-time data and control options (see Figure 2). All interaction with the XR features is done through touch in the mobile device screen. Figure 3 represents the main modules of the architecture selected to handle the scenario considered.

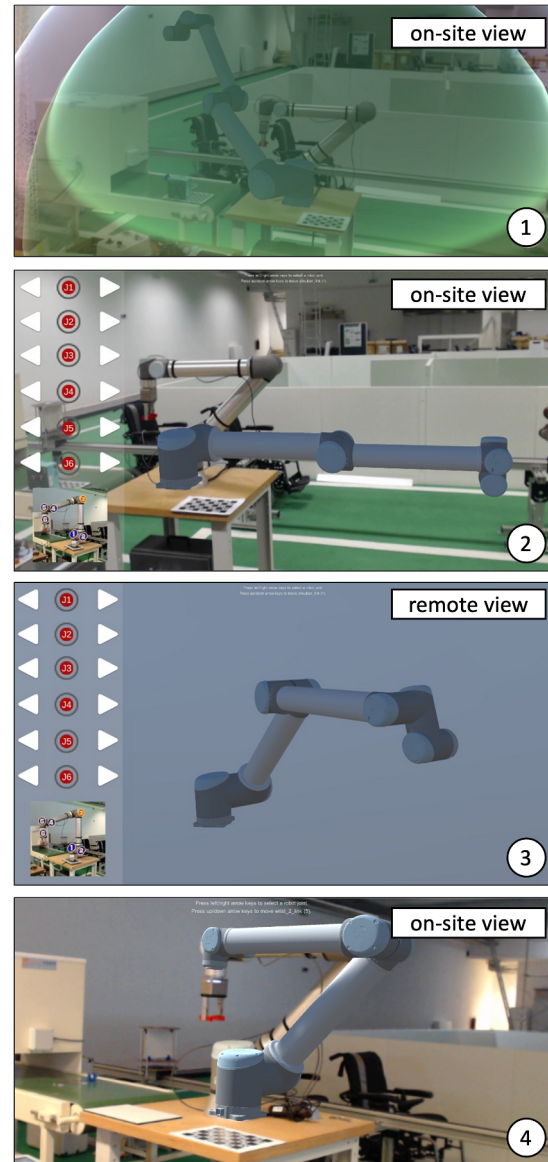


Figure 2: Illustration of the main features of the XR platform: 1- safety zones associated with the robotic arm; 2- interface used to control the robotic arm in the mobile device; 3- remote member interface including the virtual replica representation and the context stream; 4- virtual replica overlaid with the real-world robotic arm after an alteration by the remote member is performed

This also enables the safety zone feature, which plays a crucial role in ensuring on-site safety. This feature includes the display of visual spheres in distinct colors to indicate proximity to the robotic arm, providing real-time visual cues to on-site members. Furthermore, it incorporates auditory alerts, such as sound notifications, that

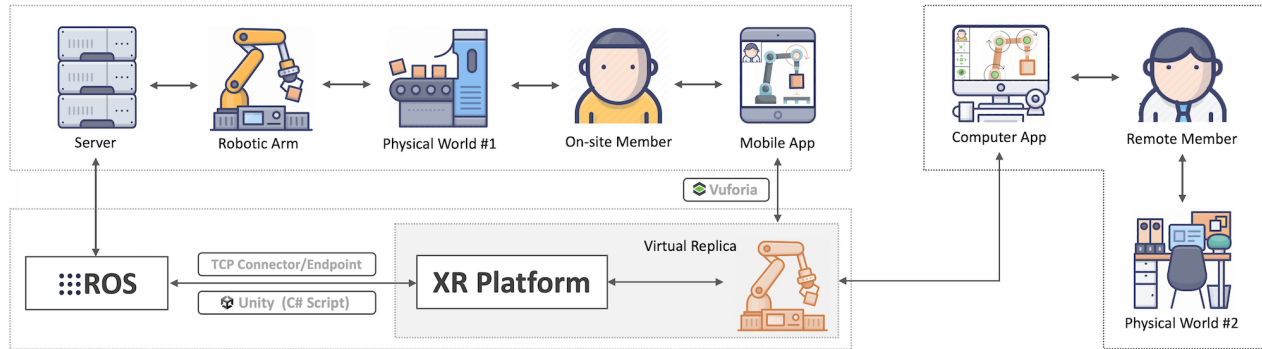


Figure 3: Architecture of the XR platform, illustrating the elements for remote collaboration between an on-site and a remote member supported by a robotic arm and XR. The different versions of the XR platform communicate with the virtual replica through the use of a TCP connector, which in turn updates the state of the physical arm. As for the XR content, it is superimposed on the real-world environment using the Vuforia library. Assets obtained from iconfinder.com

activate when a breach of these safety zones occurs. These combined visual and auditory safeguards help on-site members manage the robotic arm from a safe distance, aiming to help reduce the risk of injuries and enhancing operational safety across various environments. By providing on-site members with information directly within their field of view, we argue that XR may reduce the need for them to switch between remote screens, which are traditional used to control robotic arms. This streamlined access to information can help them make informed decisions, possibly leading to a better situation understanding.

Regarding context sharing, the on-site member uses the handheld device's camera to capture key areas of interest, which are then shared with the remote member operating from a computer. This approach establishes a common ground, enabling real-time streaming of audio and video feeds that may also include representations of collaborators, ensuring team members can effectively share perspectives and engage in discussions. Alternatively, a live stream can be obtained via a camera mounted on the robotic arm's claw, improving situational awareness for the remote member while reducing the on-site operator's burden of managing environmental views.

The remote member has access to a virtual replica of the arm, enabling remote control to adjust its position as necessary for participating in the shared task. All interactions are conducted via mouse and keyboard, using two distinct modes. The first mode employs a traditional user interface displayed on the screen, showcasing arrows for each joint of the robotic arm, allowing precise control. The second mode enables a more intuitive approach, where the remote member can hold and drag the virtual robot claw in a free mode, letting the system autonomously calculate and execute the necessary movements to reach the intended position. If required, on-site members also have the ability to control the robotic arm. In addition to the two remote control modes, they have access to a third option: pressing situational arrows overlaid directly onto the physical engines of the robotic arm. This mode offers immediate and hands-on control, enhancing flexibility and adaptability during collaborative tasks.

Either way, this is an asynchronous process, meaning that the on-site or remote member must first visualize the final position of the robotic arm and confirm the desired positioning before proceeding. Only after this confirmation is the information transmitted through the system architecture, prompting the robotic arm to execute the necessary movements. This approach ensures precision and minimizes errors, allowing for careful adjustments and thorough verification before taking action in the physical environment. Additionally, a list of the robot's available sensors can be accessed and toggled on or

off as needed, providing flexibility in data collection. This feature further enhances adaptability, enabling the system to cater to the specific requirements of diverse tasks and scenarios.

As for the virtual replica, the robotic arm in the on-site location is connected to a server running Robot Operating System (ROS) (Figure 2), which is responsible for updating the status of the virtual replica, so that, when accessed by others, all team members have access to a synchronized information. This can be done through the different versions of the XR platform thanks to the use of the ROS# plugin. This endpoint facilitates information passing to and from Unity-based applications. The versatility of this generic architecture ensures its capacity to support the active participation of multiple team members in shared activities within a collaborative framework. Additionally, the architecture is inherently scalable, capable of supporting multiple robots, each with distinct characteristics tailored to specific functionalities or roles.

The proposed tool was developed using the Unity 3D game engine, utilizing C# scripts as the foundation for its functionality. Unity was selected for its versatility in creating XR applications and its robust capability to interface with the ROS, enabling seamless integration between virtual and physical components. Virtual content was anchored within the real-world environment through marker-based technology, leveraging the Vuforia library to ensure precise placement and interaction. For the camera attached to the robotic arm claw the Orbbec Astra 3D camera was selected for its compatibility with the ROS framework. Mounted directly onto the robot, this camera provides an independent and aligned operational perspective, further enhancing the collaboration experience.

3 NEXT STEPS

Human-to-human collaboration supported by a robot is a multidisciplinary field that demands a human-centered design perspective, combined with an engineering approach, to address its inherent complexity and create effective solutions. Building on this foundation, this work proposes an initial architecture, materialized through the aforementioned prototype, as a first step toward developing a solution that better connects distributed team members. This prototype serves as a tangible starting point, demonstrating the potential to improve collaboration across remote and on-site participants. By laying the groundwork for such a solution, the work paves the way for more effective and seamless interactions within human-robot collaborative environments.

Yet, additional work is still necessary to refine and expand the system's capabilities. While the prototype serves as a promising starting point, further development is required to enhance its scala-

bility, usability, and integration with emerging technologies. This includes improving the interaction between team members, optimizing the performance of the robotic arm, and expanding the system to support more complex environments and diverse applications. To elaborate, the next steps in this work will involve expanding the capabilities of the system through the integration of a new XR librarian (e.g., AR Core). This advancement aims to move beyond the reliance on traditional markers, and instead introduce the ability to support larger, more dynamic environments that persist over time. By enabling this extended functionality, the system will provide a more robust framework for immersive experiences, offering greater scalability and sustainability in diverse applications [10, 16].

Another important step involves integrating advanced XR devices to enhance the collaborative experience for both on-site and remote members. For on-site participants, devices like the Microsoft HoloLens 2 can provide a hands-free interface to visualize additional layers of information seamlessly within their environment, boosting situational awareness and efficiency [15, 17]. For remote members, incorporating devices such as the VIVE Focus 3 or the Oculus Quest 3 enables a more immersive experience, particularly when controlling robotic arms or interacting with virtual environments. These innovations aim to bridge the gap between physical and digital spaces, creating a more intuitive and effective collaborative framework.

Furthermore, we intend to incorporate new sensors and cameras into the robotic arm claw (such as LiDAR, thermal imaging, or air quality sensors, among others), being able to perform a more robust data collection of the task context, which may help remote members to have a more comprehensive understanding of the environment conditions. Afterward, move into a mobile setting, allowing the robotic arm to incorporate the capacity to move beyond the environment. This may be done by selecting new hardware and implementing new intelligent algorithms to enable mobile robots to make more autonomous decisions, e.g., path planning, predict collaborators intentions, and assist them more effectively during remote sessions. This may allow to reduce the cognitive load on the team members [1, 18].

We are also interested in exploring the visual and auditory representation of collaborators within XR environments to enhance the sense of presence and interaction. This includes refining avatar designs to better capture realistic facial expressions, gestures, and movements, thereby creating a more lifelike and relatable virtual presence. Additionally, advancements in voice modulation and spatial audio will be considered to ensure that communication feels natural and immersive, mirroring real-world interactions. These improvements aim to foster deeper connections and more effective collaboration in XR settings, bridging the gap between virtual and physical experiences [35, 39].

We also intend to conduct a user study exploring tasks with different levels of complexity, to verify the impact of the proposed approach. This will be performed with domain experts in XR and robotics, taking advantage of on-going collaborations with the industry sector. Finally, after having the basic infrastructure to handle remote scenarios, we aim to address multi-user and multi-robot scenarios, which will force the need to ensure synchronization, ownership, coordination, communication protocols, and task allocation and other dimensions of collaboration [14, 22].

4 FINAL REMARKS

Remote collaboration has become a cornerstone of modern work, establishing itself as an essential practice across industries worldwide. As organizations adapt to increasingly globalized operations and dynamic work environments, the ability to collaborate seamlessly across distances has become paramount. Advancements in digital communication tools and immersive technologies have enabled distributed teams to work together efficiently, breaking down geo-

graphical barriers and fostering innovation. This shift is redefining how tasks are approached and executed, highlighting the importance of connectivity, adaptability, and the need for continuous improvement in remote solutions.

The integration of a robotic arm within an XR environment represents a transformative step in redefining remote collaboration practices. This combination addresses persistent challenges in remote interaction with on-site physical objects. It provides the means for remote members to engage more effectively with the physical environment. Plus, it has the potential to significantly extend the remote member's perspective, enabling a more comprehensive view of the task at hand and overcoming the limitations inherent in relying solely on the input captured by on-site team members.

The adopted architecture integrates human and robotic elements, showcasing an approach to facilitate shared activities, laying the groundwork for enhanced collaboration in distributed scenarios. The XR platform, with its features encompassing streaming the task context, on-site manipulation of the robotic arm, as well as remote control capabilities, and the visualization of safety zones, among others, underscores the potential for immersive and real-time collaboration.

Future endeavors must prioritize developing robust annotation features that cater to both on-site and remote participants, fostering clearer and more effective communication. Additionally, integrating mobile robots equipped with robotic arms can significantly augment remote expert capabilities, offering a personal perspective and greater freedom to navigate and explore the environment. This setup not only enhances situational awareness but also allows remote experts to provide hands-on assistance, revolutionizing how tasks are managed in distributed settings.

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